

DEFINITION — HOW DATA IS DESCRIBED

API INTERFACES6

 **ODCS**
BITOL @ LF

 **OpenAPI**
LF

 **AsyncAPI**
LF

 **GraphQL**
LF

 **gRPC**
CNCF

 **OData**
OASIS

DATA PRODUCTS4

 **ODPS**
BITOL @ LF

 **DPDS**
ODM

 **ODPSpec**
LF

 **DPROD**
OMG

SCHEMA5

 **XML Schema**
W3C

 **JSON Schema**
IETF

 **SQL DDL**
ISO/IEC

 **AVRO Schema**
ASF

 **Protobuf**
GOOGLE

SEMANTICS6

 **OSI**
OSI INITIATIVE

 **RDF/OWL**
W3C

 **DCAT**
W3C

 **SKOS**
W3C

 **SHACL**
W3C

 **JSON-LD**
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS6

 **CSV**
IETF

 **JSON**
IETF

 **XML**
W3C

 **PARQUET**
ASF

 **AVRO**
ASF

 **ORC**
ASF

OPEN TABLE FORMATS3

 **Iceberg**
ASF

 **Delta**
LF

 **Hudi**
ASF

STORAGE SYSTEMS2

 **S3**
AWS

 **HDFS**
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY3

 **JDBC**
JCP

 **ODBC**
ISO/IEC

 **ADBC**
ASF

INTERCONNECTION3

 **HTTP**
IETF

 **FTP / SFTP**
IETF

 **Delta Sharing**
LF

MESSAGING4

 **Kafka**
ASF

 **MQTT**
OASIS

 **CloudEvents**
CNCF

 **AMQP**
OASIS

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

 **DataFrame API**
DATA-APIS

 **Apache Arrow**
ASF

PROCESSING6

 **dbt**
DBT LABS

 **Spark**
ASF

 **Beam**
ASF

 **Pandas**
NUMFOCUS

 **Ibis**
IBIS

 **SQL DML**
ISO/IEC

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS4

 **Iceberg Catalog**
ASF

 **Unity Catalog**
LF

 **Hive Metastore**
ASF

 **Schema Registry**
CONFLUENT

LINEAGE1

 **OpenLineage**
LF

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY2

 **SQL**
ISO/IEC

 **SPARQL**
W3C

DATA QUALITY3

 **dbt**
DBT LABS

 **Great Expectations**
GX LABS

 **SodaCL**
SODA

OBSERVABILITY2

 **Open Telemetry**
CNCF

 **OORS**
BITOL @ LF

POLICIES1

 **OPA**
CNCF

AI INTERFACES2

 **MCP**
LF

 **A2A**
LF